ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Pre-tutorial 4

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This tutorial covers selected content from Chapter 2 of Murray R. Spiegel's *Applied Differential Equations*.

Note (Instructions). Work through the problem before your tutorial and upload your solution to onQ as a single PDF by 9:00 AM on the due date (no in-class submission). Show your working, not just the final answer: a genuine, complete attempt earns the 2 marks for this pre-tutorial; a missing or incomplete attempt earns 0.

Problem

Context. Week 4 notes, §3 "Making an inexact ODE exact by multiplying with an integrating factor". There the equation

$$y\,\mathrm{d}t + (3 + 3t - y)\,\mathrm{d}y = 0 \qquad \text{((12) of the Week 4 notes)}$$

is shown *not* to be exact; Spiegel's Theorem 2 supplies the integrating factor $\mu = y^2$, and multiplying by it gives Eq. (13) below, which *is* exact. The notes stop there. This pre-tutorial asks you to finish the job.

Problem 1 (Solve Eq. (13) of the Week 4 notes). Regarding the last page of the posted notes *Making inexact ODE exact by multiplying with an integrating factor*, find the solution of Eq. (13),

$$y^3\,\mathrm{d}t + y^2\,(3 + 3t - y)\,\mathrm{d}y = 0. \qquad \text{(T4.1)}$$